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ITAI 4370: AI 5/6G Comm & ORAN Net
Laboratory 02, Activity 1: Network Protocol Analysis with Wireshark
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Objective

Capture and analyze network packets to understand TCP/IP, addressing, and protocol behavior.

Capture Summary

The capture was taken on the live Wi-Fi interface (Wi-Fi: en0) and recorded 2599 packets during about one minute of normal web browsing. Three display filters were then applied to isolate traffic by type: ip, tcp, and http. The three filtered views are shown below.

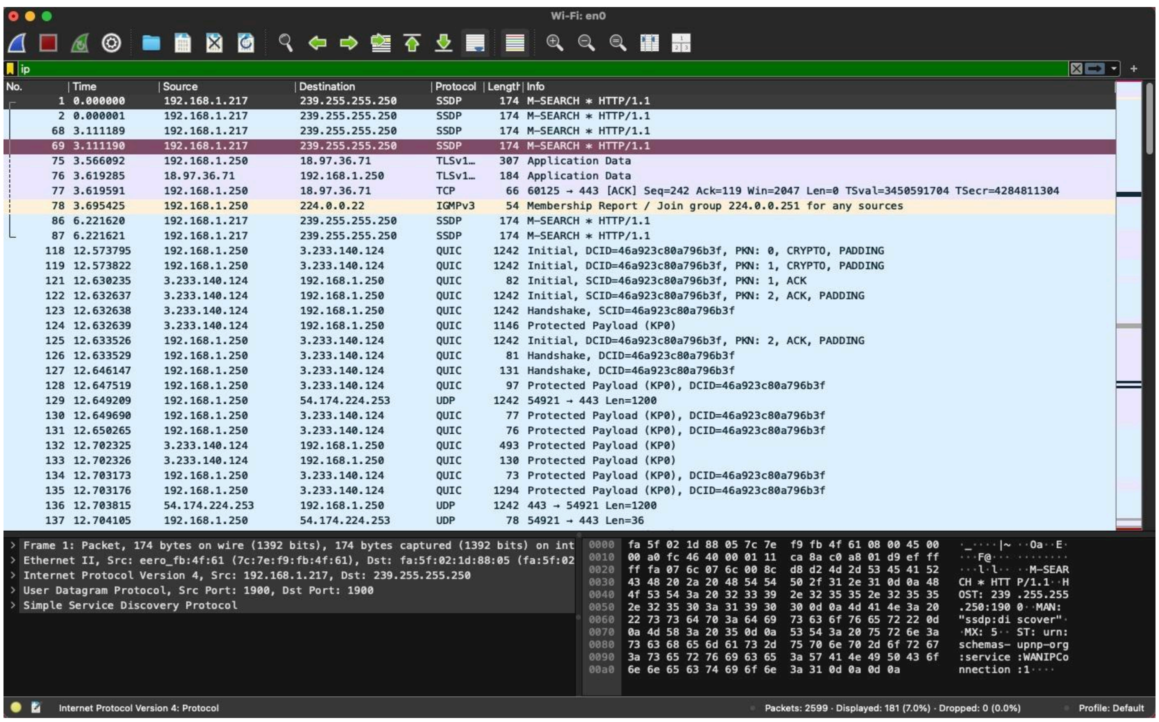

Figure 1. Capture filtered with the ip display filter (181 of 2599 packets shown).

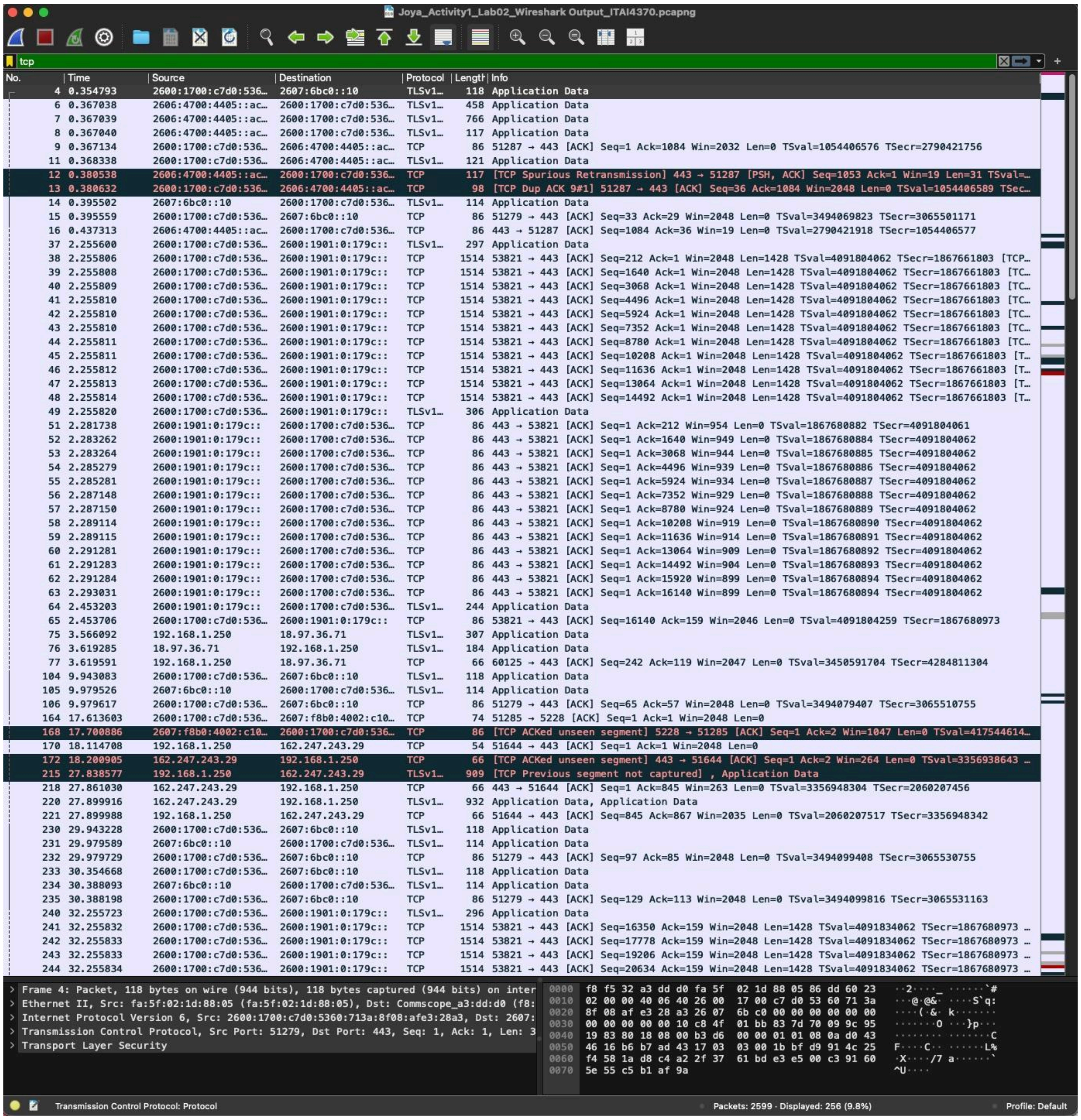

Figure 2. Capture filtered with the tcp display filter (256 of 2599 packets shown). The title bar shows the saved capture file.

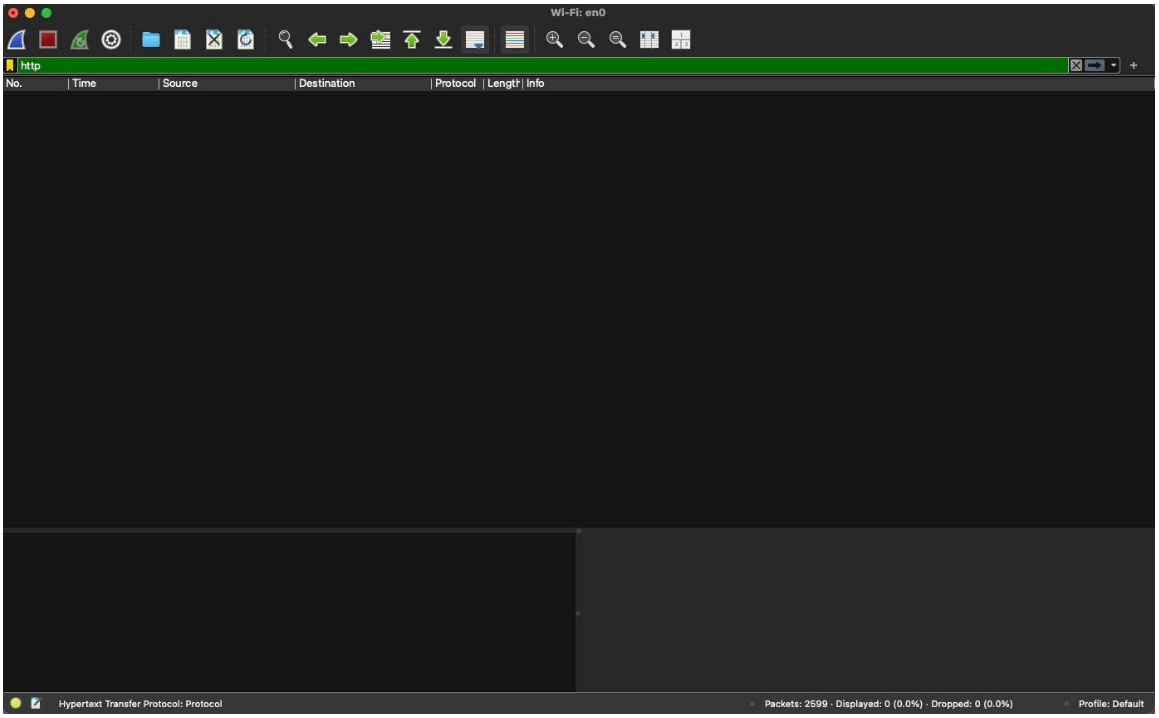

Figure 3. Capture filtered with the http display filter (0 packets shown), confirming that all web traffic in the session was encrypted.

Analysis of the Packet Headers

Source and Destination IPs. The capture shows a clear split between private local addresses and public internet addresses, and it carries both IPv4 and IPv6. On the local side, the devices on the home network appear as 192.168.1.217 and 192.168.1.250, which sit in the private 192.168.x.x range. The IPv6 view, seen under the tcp filter, shows the local device as a global IPv6 address beginning 2600:1700:c7d0. On the remote side, the traffic reaches public servers over IPv4 (18.97.36.71, 3.233.140.124, and 54.174.224.253) and over IPv6 (2607:6bc0::10 and 2600:1901:0:179c). A few destinations are multicast rather than a single host, such as 239.255.255.250 used by SSDP and 224.0.0.x used by IGMP, which devices use to reach a group on the local network instead of one machine.

Protocols used (TCP, UDP, ICMP). The capture is dominated by TCP and UDP. TCP carries the reliable, connection-based traffic, seen under the tcp filter as streams of acknowledgment (ACK) segments to and from port 443. It also includes a TCP Spurious Retransmission and a TCP Duplicate ACK, which show TCP detecting and recovering from a delayed or duplicated segment. UDP carries the fast, connectionless traffic, including QUIC, the modern encrypted web transport that also uses port 443, and SSDP device discovery. The encrypted application data appears as TLS. Classic IPv4 ICMP, the protocol behind ping, did not appear, which is expected because this was ordinary web browsing with no ping or traceroute run. The control messaging that did appear was IGMPv3 group membership, visible under the ip filter, and the full saved capture also recorded ICMPv6 neighbor discovery, which is the IPv6 equivalent.

Port numbers. The ports follow the normal client and server pattern. The local device uses high-numbered ephemeral (temporary) source ports that the operating system assigns for each connection, such as 60125, 54921, 51287, 51279, and 53821. The destination port is almost always 443, which is HTTPS, the standard port for encrypted web traffic. The one clear exception is SSDP, which uses port 1900 for both source and destination, seen in the header detail pane. Port 80, plain unencrypted HTTP, never appears, which matches the empty http filter result.

Round Trip Time (RTT). RTT is the time between a segment and the acknowledgment that confirms it. Wireshark calculates this for TCP and reports it in the SEQ/ACK analysis of a packet that acknowledges data.

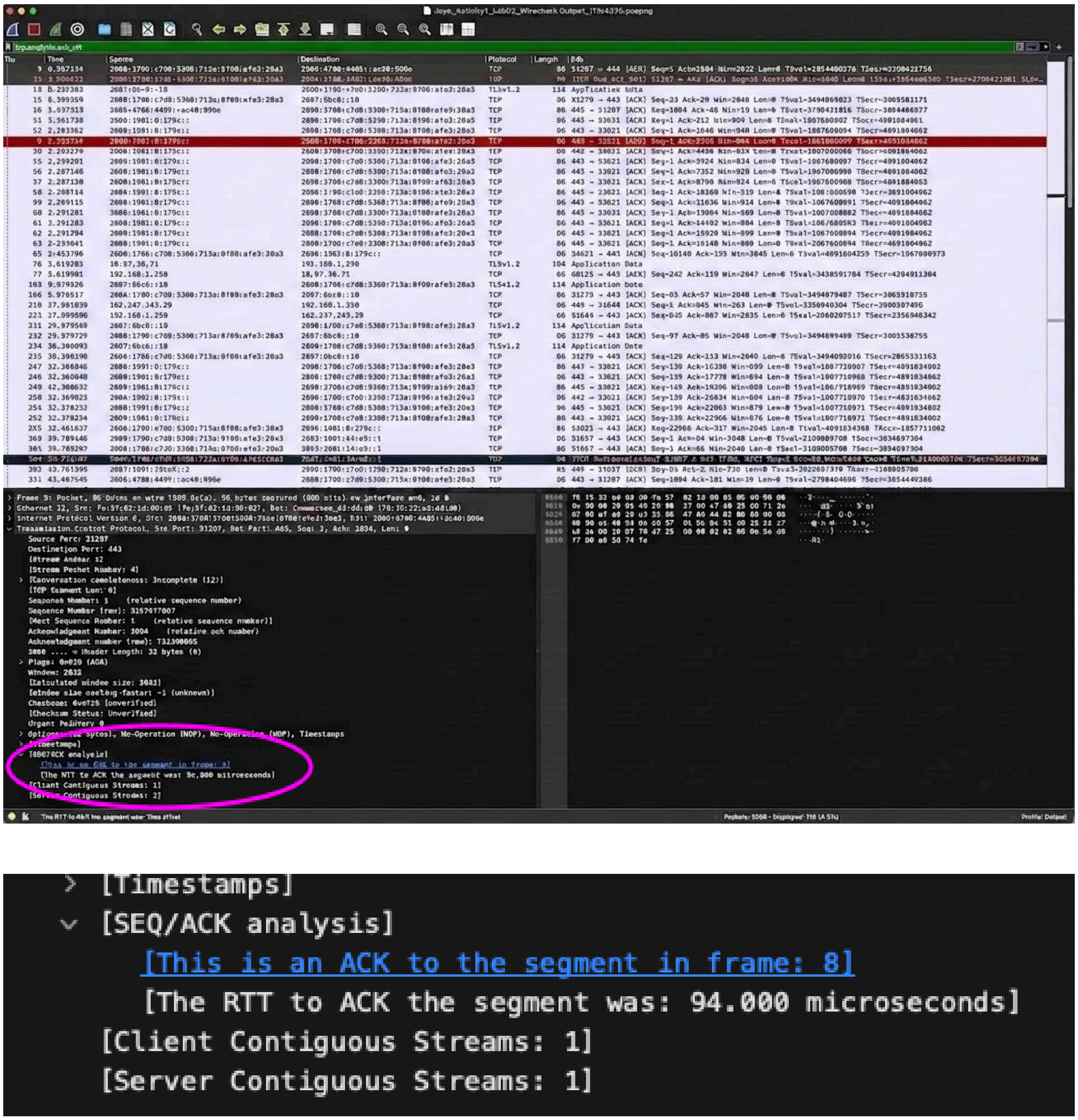

Figure 4. The RTT in the [SEQ/ACK analysis] of packet 9 under the tcp.analysis.ack_rtt filter, reading 94.000 microseconds.

Measured RTT from this capture: 94 microseconds (0.000094 seconds), from packet 9 under the tcp.analysis.ack_rtt filter. Packet 9 is the local machine acknowledging data it received, so this figure reflects how quickly that acknowledgement was sent at the capture point.
